

Foundation of Ai & Automation

ASSIGNMENT – UNIT 1

Instruction: Answer all the questions. Write clear, concise and well-structured answers.

Q. No.	Questions
1.	Define the following terms in your own words: (a) Intelligence (b) Artificial Intelligence (c) Agent (d) Rationality (e) Logical Reasoning
2.	Explain the Turing Test. What is its purpose and how does it work? Also mention one limitation of this test.
3.	Differentiate between Automation and Artificial Intelligence with the help of suitable examples. Is a calculator an AI system? Justify your answer.
4.	Consider the following situation: A person immediately pulls their hand away after touching a hot stove. Is this action rational? Is it intelligent? Explain the difference between rational behavior and intelligent behavior using this example.
5.	To what extent are the following computer systems instances of AI or simple automation? Justify your answer in each case. (a) Supermarket barcode scanners (b) Web search engines (c) Voice-activated telephone menus (d) Internet routing algorithms
6.	Why would evolution tend to result in systems that act rationally? What goals are such systems designed to achieve?
7.	What is Problem Reduction in Artificial Intelligence? Explain how a complex problem can be divided into smaller sub-problems.
8.	Explain State Space Search in AI. What is a state, initial state, goal state, and operator? Illustrate the concept with a suitable example.
9.	Explain the problem-solving algorithm used in Artificial Intelligence. Describe the major steps involved in solving a problem using search.
10.	Discuss the major components of Artificial Intelligence. Explain the role of knowledge representation, reasoning, learning, perception, and action.
